Enhancing Web Agents with a Hierarchical Memory Tree

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Paper version: arXiv 2603.07024v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *The high-level user intents are transferable, but the specific action details are not.*

Flat memory entangles **task logic** with **site-specific details** such as element IDs.

Motivation: three failure modes

1. **Entanglement.** Intent is paired with invalid element IDs.
2. **Workflow mismatch.** Retrieved steps skip or repeat stages.
3. **Context pollution.** Raw HTML trajectories are long.

Evaluation setup

Mind2Web · offline | WebArena · online | 3 splits: task / website / domain | GPT-4, temperature 0 | K = 5 / 8 / 5 retrieval

- **Offline:** Mind2Web step SR. **Online:** WebArena task SR; kept apart.
- **Not stated:** seeds, repeats. Online memory starts empty.

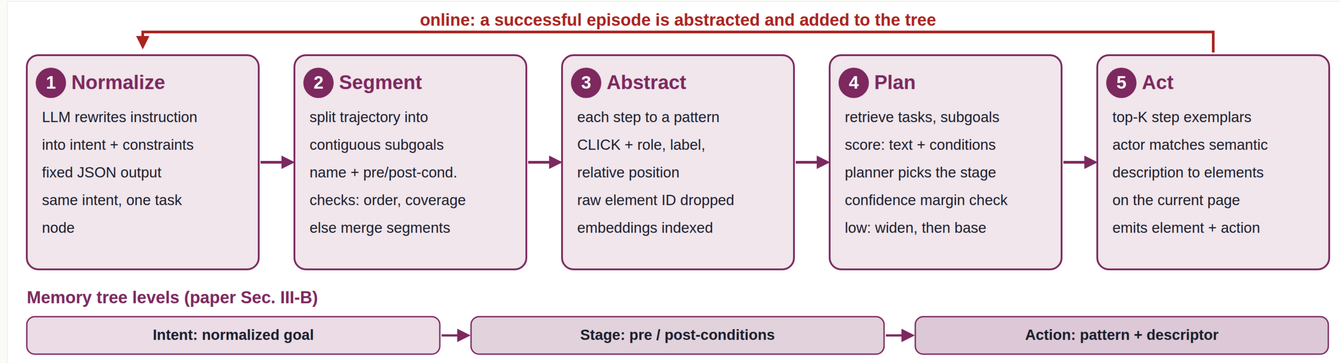
Our read (CUA Radar)

66	★★	online
score of 100	band	evidence level

- **Reports:** online WebArena eval; cross-website split; AWM and flat baselines; ablation; cost.
- **Flags:** AP5, data overlap. **Not reported:** variance, matched budgets, code.
- Pattern K27. Authors' numbers, not rerun.

Method: a three-level memory tree, a stage-aware planner and an actor

- **Sees / acts:** simplified DOM or a11y tree in; CLICK / TYPE on elements out.
- **Memory:** intent, stage (pre/post-conditions) and action levels; no raw IDs stored.
- **Signal:** successful trajectories only, abstracted by an LLM; no training.



Our redraw (not the paper's figure): construction pipeline, stage-aware inference and the three memory levels.

$$\mathrm{Score}(g) = (1-\lambda)\cos\big(q_G, \mathrm{emb}(g)\big) + \lambda\,\mathrm{CondMatch}(g, o_t)$$

Subgoal score: embedding match plus condition match ($\lambda = 0.3$)

Key results: WebArena task success (%), online

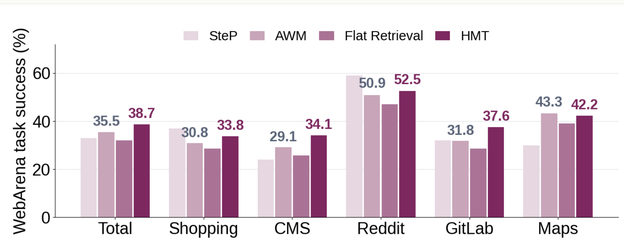


Table II, redrawn: GPT-4 backbone; total is a weighted average; no std reported.

HMT 38.7 vs AWM 35.5; Flat Retrieval 32.1

SO WHAT → *Reddit: SteP 59.0 is above HMT 52.5. Maps: AWM 43.3 is above HMT 42.2.*

Ablation: WebArena total task success (%)

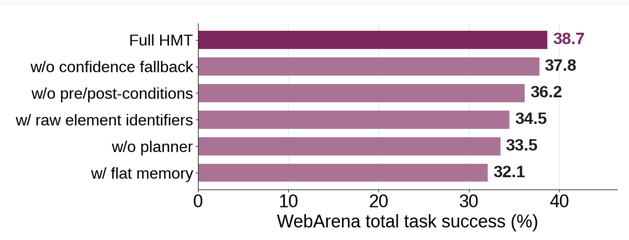


Table III, redrawn: one component removed or replaced at a time.

SO WHAT → *Flat memory (32.1) and no planner (33.5) cost most online. Raw identifiers cost most offline: Mind2Web cross-website step SR falls from 39.7 to 12.4. Grounding success on cross-website is 76.8% with semantic descriptions against 12.4% with raw identifiers. On cross-task, where DOM structure is familiar, raw identifiers still ground 92.1% and HMT is comparable to AWM.*

Headline Numbers

38.7%
WebArena task success
vs 35.5% AWM, 32.1% flat

- **39.7%** Mind2Web x-site step SR
- **84.2%** Recall@5, cross-website
- **5.2** avg steps per task

The paper's own controls

CONTROL	HMT	COMPARED WITH
Mind2Web x-site step SR	39.7	AWM 33.7 / 33.9
Mind2Web x-task step SR	48.5	AWM 45.1 / 43.6
Recall@5, cross-website	84.2%	flat 65.8%
Context tokens, $ / task	3,120, 0.11	flat 11,450, 0.38

Takeaway

A three-level memory with semantic element descriptions lifts WebArena to 38.7% and Mind2Web cross-website step SR to 39.7%; gains are small on cross-task and in Maps.

GPT-4 backbone | 3,120 vs 11,450 tokens | no std / seeds reported

Scan to read the full paper

arxiv.org/abs/2603.07024v1